PERSONALIZED TEXT GENERATION

# StyleVector

Contrastive activation steering: reproduction, ablations and extensions

**Slok Tulsyan · 12342080**

02 · PROBLEM

# Problem statement

### One average style

- LLMs write in one generic voice
- Real user needs a person’s own style

### RAG

- Injects past texts into the prompt
- Latency grows with history
- Mixes content with style

### PEFT

- Trains an adapter per user
- Per-user training and storage

**StyleVector: one vector per user in activation space. No training, no retrieval.**

# Three-stage pipeline

1

## Style-agnostic response

A general LLM rewrites each history input without personal style

$$\hat{y}_i = M_g(x_i)$$

2

## Contrastive activations

Last-token activation of the real response (a_p) minus the generic one (a_n), averaged over the history, at one layer

$$s = \frac{1}{|P_u|} \sum_i (a_{p,i} - a_{n,i})$$

3

## Additive steering

Add the scaled vector at every generated token

$$h' = h + \alpha s$$

Setup: LLaMA-2-7B-chat (4-bit), LaMP-4 headlines, LaMP-5 scholarly titles, LaMP-7 tweet paraphrasing, greedy decoding, ROUGE-L and METEOR, 1,000 dev users per task, history cap 15.

# Steering strength and sample size

### α sweep, one user (Qwen2.5-3B, layer 20)

- α ≤ 0.50: identical to baseline
- α 0.75 to 1.50: changed, coherent
- α 2.0: degraded text
- α 4.0: collapsed, Mean Diff 0.000

### Sample size changes the verdict

- n = 10: 3 of 6 cells improve
- n = 50, LaMP-4: +19.4% ROUGE-L
- n = 50, LaMP-7: −38.9% ROUGE-L
- Small model unstable, so moved to LLaMA-2-7B-chat

First PCA check (n = 5, α = 1.0, LaMP-5): top-3 PCA rescaled to the mean-difference norm gave +18.6% ROUGE-L, +11.4% METEOR.

# ROUGE-L gain (%) by iteration

| Configuration | LaMP-4 | LaMP-5 | LaMP-7 |
|---|---|---|---|
| Paper | +3.3 | +25.8 | +12.8 |
| Layer/α grid search, n≈50 | +3.1 | −4.4 | +24.8 |
| Shuffled dev split, n≈201 | +1.6 | +2.6 | +29.2 |
| One prompt variant for all tasks, n≈200 | −0.4 | +6.2 | −9.4 |
| Prompt variant per task, n≈200 | −5.6 | −16.4 | +29.2 |
| Larger sample, n≈800 | −2.2 | −2.2 | +26.4 |
| Robust grid search, n≈800 | −4.5 | +0.6 | +26.4 |
| All dev users, n≈806 | −0.8 | +0.6 | +26.4 |
| Clean setup, n = 300 | +4.8 | +29.8 | +16.4 |
| Clean setup, n = 1,000 | +3.3 | +23.1 | +16.5 |

Selected layer / α: 16 / 0.25 (LaMP-4), 16 / 0.50 (LaMP-5, LaMP-7). Grid: layers 16 to 28, α 0.25 to 3.0.

06 · DIAGNOSTICS

# Diagnosing the gaps

## Output format

- 97% (LaMP-4) and 100% (LaMP-5) of baseline outputs start with a preamble
- 22 to 24 words against 9 to 10 in the references
- Fixed in the clean setup

## Ceiling effect

- corr(baseline, gain): −0.36 (LaMP-4), −0.28 (LaMP-5)
- Weak-baseline users gain +0.009 / +0.024
- Strong-baseline users change −0.019 / −0.022

## Other checks

- History items over 512 tokens: 0 to 0.3%
- LaMP-4 α 0.05 to 0.25: within ±3%
- History cap 60: −0.8%, +0.2%
- LongLaMP Topic Writing (n = 38): −6.5%

07 · RESULT

# StyleVector reproduced

**6 / 6**
task–metric pairs improved

**+23.1%**
LaMP-5 ROUGE-L (paper +25.8%)

**+16.9%**
LaMP-7 METEOR (paper +17.5%)

| Task | Metric | Paper base | Paper steered | Paper Δ | Ours base | Ours steered | Ours Δ |
|---|---|---|---|---|---|---|---|
| LaMP-4 | ROUGE-L | 0.0398 | 0.0411 | +3.3% | 0.0988 | 0.1020 | +3.3% |
| LaMP-4 | METEOR | 0.0790 | 0.0809 | +2.4% | 0.1172 | 0.1207 | +3.0% |
| LaMP-5 | ROUGE-L | 0.1086 | 0.1366 | +25.8% | 0.1708 | 0.2102 | +23.1% |
| LaMP-5 | METEOR | 0.2337 | 0.2575 | +10.2% | 0.2829 | 0.3125 | +10.5% |
| LaMP-7 | ROUGE-L | 0.2506 | 0.2827 | +12.8% | 0.2525 | 0.2942 | +16.5% |
| LaMP-7 | METEOR | 0.2588 | 0.3042 | +17.5% | 0.2770 | 0.3239 | +16.9% |

08 · EXTENSION

# Two modifications to the method

### Multi-component vector

- Top-k PCA of the contrastive differences
- Weighted combination of components
- Targets single-vector conflation of style

### Norm-preserving steering

- Rescale to the original norm after adding
- $h' = ||h|| \cdot (h + p) / ||h + p||$
- Targets collapse at large $\alpha$

| | Additive | Norm-preserving |
|---|---|---|
| Single vector | Paper’s method | Ablation A |
| Multi-component | Ablation B | Full proposed method |

# Five runs of the extension

| Run | What changed | Users per task |
|---|---|---|
| 1 | Fixed settings, raw outputs, five conditions | 300 |
| 2 | Validation-tuned layer and strength, cleaned outputs, global-vector and shrinkage controls | 150 |
| 3 | 60 validation users, wider strength grid (to 1.2), random-direction control | 300 |
| 4 | k = 1, 3, 5; layer and strength sweeps | 300 (sweeps 50) |
| 5 | Refined pipeline: cleaned outputs, repaired tweet inputs, per-condition tuning, shared-basis multi-component vector | 300 |

# What Runs 1 to 4 show

**14–51%**
baseline ROUGE-L gain from output cleaning alone

**24 of 24**
tuned strengths at the grid minimum

**2 of 42**
norm-preservation comparisons significantly better

- Paper's method vs. no steering (18 comparisons): 3 better, 1 worse, 14 inconclusive
- Multi-PCA vs. single vector: 1 better, 1 worse of 18
- LaMP-7 gain: +29% on raw output, +12.5% after cleaning
- Global vector ≈ user vector (Run 2); random direction ≈ PCA + NP (Run 3)
- At 4× strength on LaMP-7 outputs collapse; norm preservation does not prevent it

# Refined pipeline

| Condition | L4 R-L | L4 MET | L5 R-L | L5 MET | L7 R-L | L7 MET |
|---|---|---|---|---|---|---|
| No steering | 0.0569 | 0.0574 | 0.2578 | 0.3076 | 0.3001 | 0.4055 |
| Mean Diff (paper) | 0.0609 | 0.1436 | 0.3141 | 0.3411 | 0.3527 | 0.4147 |
| Mean Diff + NP | 0.0676 | 0.0844 | 0.3100 | 0.3425 | 0.2989 | 0.3620 |
| Per-user PCA, k = 3 | 0.0636 | 0.1590 | 0.3178 | 0.3392 | 0.3522 | 0.4148 |
| Shared basis | 0.0553 | 0.1341 | 0.3019 | 0.3321 | 0.3512 | 0.4121 |
| Shared basis + NP | 0.0679 | 0.0905 | 0.3036 | 0.3345 | 0.2931 | 0.3665 |

- Shared basis vs. Mean Diff: no significant difference in 6 of 6 comparisons
- Norm-preserving variants lower than Mean Diff in 3 of 6 comparisons each; selected $\alpha \leq 1.0$
- Per-user PCA matches Mean Diff, better on LaMP-4 METEOR (+0.0155)

# Conclusions

### Reproduction

- Improves on 6 of 6 task–metric pairs at n = 1,000
- Direction and size match the paper

### Extensions

- Shared basis equals Mean Difference
- Norm preservation lower in half of comparisons
- Per-user PCA matches Mean Difference

### Practice

- Layer and strength tuning matter
- Output cleaning: 14–51% effect
- Controls show what is personal

**Mean Difference steering with tuned layer and strength is the strongest configuration tested.**